years of building and measuring actual hardware. Anyone claiming otherwise is either mistaken or promoting an unrealistic view of the technology.

What they are genuinely good at is a different and complementary set of tasks. They are particularly effective at reviewing large bodies of technical literature and surfacing relevant precedents. Ask a well-prompted LLM to identify prior art on impedance matching techniques for dual-band handset antennas, and it can produce a structured, well-referenced survey in minutes rather than hours. It will not be perfect—hallucinations are a real risk, and all references must be verified—but the starting point it provides is significantly better than starting from scratch.

They are also effective at helping engineers think through the logical structure of a design problem. The process of articulating a constraint or a trade-off clearly enough for an AI to understand often helps engineers understand it more clearly themselves. Several RF engineers interviewed for this article described using LLM conversations as a structured form of rubber-duck debugging—the act of explanation forces clarity.

Practical applications in antenna design

Specific use cases are emerging with enough consistency to be worth documenting. Antenna engineers are using LLMs to generate initial parameter sweeps for common antenna geometries, dipoles, patches, slots, and PIFA structures, using established analytical models. The output is not a finished design but a reasonable starting point for simulation that reduces several rounds of manual calculation.

Script generation for simulation automation is another area where LLMs are proving their value. Tools like HFSS, CST, and FEKO all have scripting interfaces, but the documentation is dense, and the syntax is unforgiving. LLMs trained on large code corpora have absorbed enough of this material to generate functional scripts for common tasks—geometry parameterisation, frequency sweep setup, and field export—reliably enough for many engineers to incorporate them into their workflows.

Post-processing of simulation data is a third application. Converting raw S-parameter data into insights, identifying resonances, quantifying bandwidth, and assessing efficiency trends across a parameter space involves a pattern recognition task that LLMs, when given well-structured data and clear instructions, handle competently. The engineer still makes the final interpretation and decisions; the AI simply accelerates the data processing that supports them.

The risks engineers need to understand

Responsible use of LLMs in RF design requires clear-eyed awareness of their failure modes. The most dangerous is 'confident incorrectness,' the tendency to state incorrect information in the same authoritative tone used for correct information. An LLM that gives the characteristic impedance of a microstrip line with the same level of confidence as it gives the speed of light exhibits a behaviour that has caused real problems in engineering contexts. Every numerical output, every cited reference, and every specific technical claim must be independently verified.

The second significant risk is the homogenisation of design thinking. If every engineer in a company is using the same AI tool trained on the same body of literature, the diversity of approaches that drives genuine innovation may gradually erode. LLMs reflect the central tendency of their training data; they are, by construction, conservative. Counterintuitive designs and approaches that challenge conventional wisdom in productive ways are less likely to emerge from an AI-assisted workflow than from engineers willing to think against the grain.

Getting started without getting misled

For RF engineers who want to begin experimenting with LLMs in their workflow, a few practical principles apply.

First, use LLMs for tasks where the output can be immediately verified—literature searches, script generation, and document summarisation—before relying on them for tasks where verification is more difficult. Build familiarity with the tool's reliability in low-stakes contexts before deploying it in high-stakes ones.

Second, prompt specificity matters enormously. The difference between "help me design an antenna" and "I need to match a 50-ohm source to a patch antenna resonating at 2.45GHz on an FR4 substrate 1.6mm thick; suggest an inset feed approach and estimate the inset depth analytically" is the difference between a generic response and a useful one. LLMs reward engineers who can articulate their problems precisely, which is itself a core engineering skill.

Third, maintain technical competence. The greatest risk of any productivity tool is that it atrophies the skills it replaces. LLMs should be used to move faster through tasks that are already understood, not to bypass understanding of unfamiliar ones. When the tool fails—and it will fail—engineers must be capable of recognising and correcting the failure.

The RF design lab is changing. LLMs are part of that change, alongside better simulation tools, improved measurement hardware, and the growing body of technical literature on which these models are trained. The engineers most likely to benefit are those who approach these tools with the same curiosity and scepticism they apply to any new instrument on the bench. EFY